LoRA Ensembles

LoRA Ensembles[106]

LoRA Ensembles replaces the costly multi-model training in traditional Deep Ensembles with multiple Low-Rank Adapters (LoRA) attached to a shared base pre-trained model

FineTune Procedure

- Each adapter learns a low-rank update to the frozen base model weights:

$$\Delta W = \alpha BA \qquad B \in \mathbb{R}^{d\times r} \quad A \in \mathbb{R}^{r\times k}$$

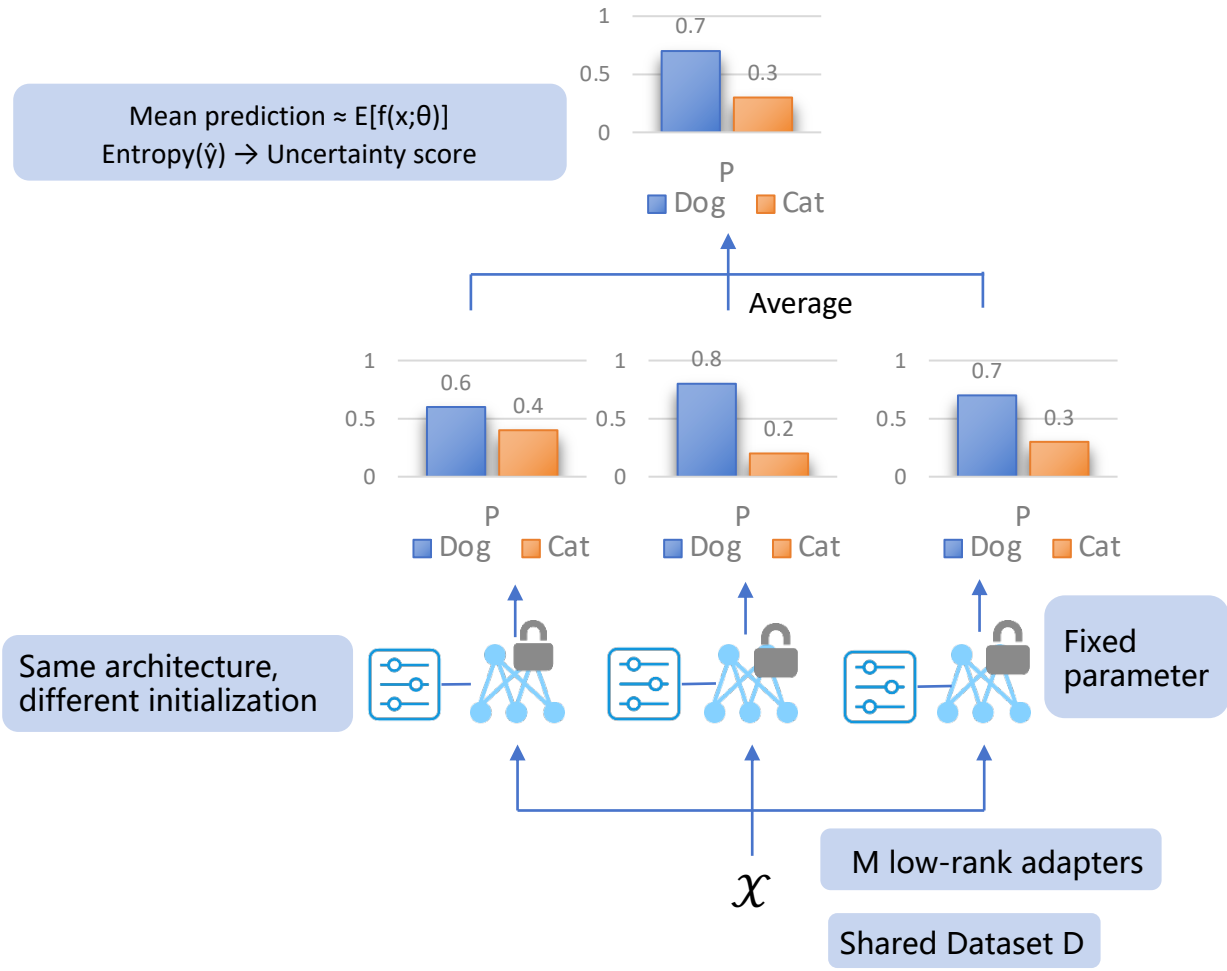
$$L(\Delta W) = \min_{\Delta W} \sum_{n} -\log p(y_n|x_n; W^* + \Delta W)$$

- Initialize A randomly, $\mathrm{B} = 0$;
- KL regularization, strong weight decay, early stopping

Ensemble Aggregation

- Final prediction combines M adapters:

$$p_{ens}(y|x) = \frac{1}{M}\sum_{m=1}^{M} p(y|x; W_m)$$



[106] LoRA ensembles for large language model fine-tuning, arxiv 2023

LoRA Ensembles[106]

- **Task Type:** Multiple-choice QA & Binary classification

- **Introduction:** LoRA Ensembles adapts Deep Ensembles by attaching multiple independently initialized low-rank adapters to a shared LLM backbone and averaging their token-level distributions. It quantifies parameter uncertainty and provides a parameter-efficient ensembling strategy within standard LLM fine-tuning workflows.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Ensemble	Parameter	White-box	Fine-tuning	High	Calibration

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